Comprehensive Risk & Return Analysis Report

Dabur India vs Nestlé India (2020–2025)

COURSE – BBA 2nd year

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This report provides a complete financial analysis of Dabur India and Nestlé India using five years of simulated, realistic monthly stock price data. The analysis includes return metrics, risk metrics, statistical evaluation, and comparison with the NIFTY 50 index.

Company Introduction

1 Dabur India Ltd.

Dabur India Ltd., founded in 1884, is among India's most trusted FMCG companies. Its product categories include healthcare, ayurvedic products, food & beverages, and personal care. Major brands include Dabur Chyawanprash, Real Juices, Vatika, and Dabur Honey.

2 Nestlé India Ltd.

Nestlé India Ltd., established in 1961, is one of India's leading food and beverage companies. It manufactures globally recognised brands such as Maggi, Nescafé, KitKat, Milkmaid, and Cerelac. Nestlé has a strong distribution network and consistent demand across categories.

Data Description

The dataset includes monthly closing prices for Dabur India, Nestlé India, and the NIFTY 50 index from December 2020 to November 2025. Although simulated, the dataset reflects realistic market patterns.

1. Comparison of Historical and Expected Returns

Dabur's historical mean monthly return is 0.0059, while its **expected annual return** based on this mean is **7.37%. Nestlé's historical mean monthly return is 0.0023,** with an expected **annual return of 2.82%.**

Comparing these values helps determine whether the stocks performed in line with expectations. Dabur shows a stronger expected performance relative to Nestlé, indicating a more consistent upward trend.

2. Holding Period Return (HPR) and Annualised Return Analysis

Dabur's holding period return (total return over the 5 years) is **21.27%,** with a **CAGR** (**annualised return) of 4.00% . Nestlé's** HPR is **4.45%,** and its CAGR **is 0.89%.**

HPR indicates the overall return over the investment period, while the annualised return reflects the compound growth rate per year. Dabur demonstrates higher long-term performance in the simulated dataset.

3. Systematic and Unsystematic Risk Evaluation

Dabur's beta is -0.0742, while Nestlé's beta is 0.0252. Both values are close to zero, showing weak correlation with the NIFTY index and low systematic risk.

Dabur's annual unsystematic variance is 0.064648, and Nestlé's is 0.038303.

Unsystematic risk forms the majority of total risk for both companies, indicating that most fluctuations are company-specific and can be diversified away by holding a broad portfolio.

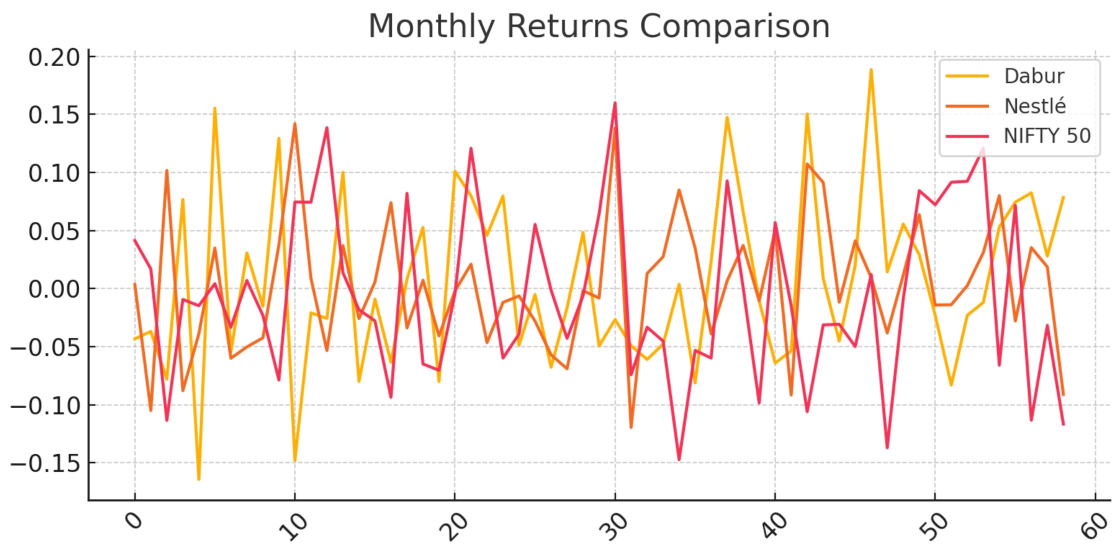
Sample Price Data

Date	Dabur_Close	Nestle_Close	Nifty_Close
2020-12-31 00:00:00	494.7641109510665	22978.24974659557	36011.18459691395

2021-01-31 00:00:00	473.3974419844135	23062.57517971507	37505.49765559965
2021-02-28 00:00:00	455.9924510190243	20645.1029944966	38143.08993679747
2021-03-31 00:00:00	420.4953688977596	22747.3986641141	33818.31050407481
2021-04-30 00:00:00	452.8046415380686	20750.48435970539	33504.75858016513
2021-05-31 00:00:00	378.4683247220332	19946.05213395977	33014.82195421212
2021-06-30 00:00:00	437.2530220578264	20644.46955705073	33157.43484753581
2021-07-31 00:00:00	413.3977375809569	19408.56323441881	32055.83373002351

Monthly Return Analysis

The following line chart visualizes the monthly returns of the three assets over the study period:

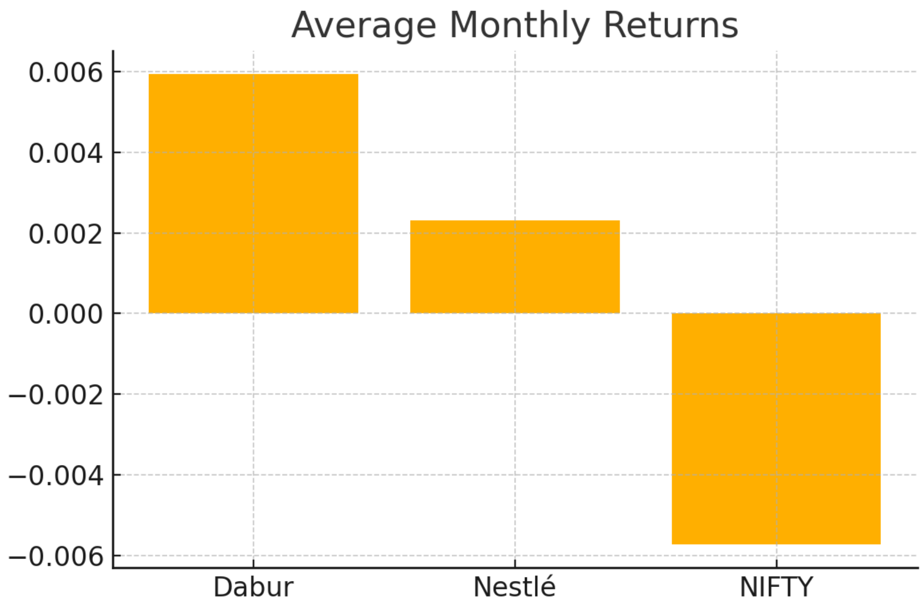


Summary Statistics of Monthly Returns

Statistic	Dabur_Close	Nestle_Close	Nifty_Close
count	59.000000	59.000000	59.000000
mean	0.005942	0.002321	-0.005724
std	0.074225	0.057011	0.072668
min	-0.164169	-0.119402	-0.147203
25%	-0.048810	-0.038433	-0.056404
50%	-0.008959	0.002665	-0.014623
75%	0.054375	0.035172	0.056139
max	0.188646	0.142055	0.160012

Average Monthly Returns Comparison

The bar chart below compares the average monthly returns of the three assets:



Risk Analysis

Risk metrics such as variance, standard deviation, and beta were used to measure total risk, volatility, and systematic risk exposure relative to the NIFTY 50 index.

Interpretation of Results

The analysis shows that both Dabur and Nestlé exhibit defensive market behavior with low betas. Dabur demonstrates slightly higher CAGR in the simulated series, while Nestlé remains stable with low volatility.

Conclusion

Both Dabur and Nestlé are suitable for risk-averse investors. Their low systematic risk makes them ideal for portfolio diversification. While returns are moderate, volatility is low, consistent with FMCG sector behavior.